Coding Theory and Mathematics

– Linear, Hamming, and Cyclic Codes –

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PQC Hard Problem Colloquium

2026 Spring

May 18, 2026

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I. Introduction

I. Introduction
Errors Are Everywhere



Physical world

- Scratches on a CD or DVD

Digital world

- Bit flips during internet transmission



Questions.

"How do we transmit (or store) data **reliably** in the presence of noise?"

I. Introduction

Can We Make Noisy Data Reliable Again?



$$\cdots 1\,1\,1\,1\,0\,1\,1\,0\,1\,0\,1\,0\,1\,1\,0\,0\,1\cdots$$

noise

$$\cdots 1\,1\,1\,1\,0\,1\,1\,0\,1\,0\,1\,0\,0\,1\,0\,0\,1\cdots$$

I. Introduction
Can We Make Noisy Data Reliable Again?



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noise

$$\cdots 1\,1\,1\,1\,0\,1\,1\,0\,1\,0\,1\,0\,0\,1\,0\,0\,1\cdots$$

- Can we **detect** it?

I. Introduction
Can We Make Noisy Data Reliable Again?



$$\cdots 1\,1\,1\,1\,0\,1\,1\,0\,1\,0\,1\,0\,1\,1\,0\,0\,1\cdots$$

noise

$$\cdots 1\,1\,1\,1\,0\,1\,1\,0\,1\,0\,1\,0\,0\,1\,0\,0\,1\cdots$$

- Can we **detect** it?
- Can we **fix** it?

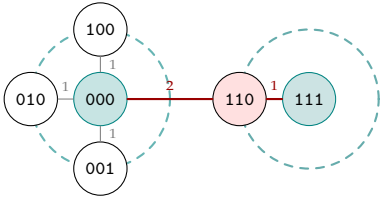
I. Introduction

Triple Repetition Code $[3, 1]_2$

The simplest way to correct errors: **say it three times**.

$$0 \longmapsto 000, \qquad 1 \longmapsto 111.$$

Received	Majority	Decision
000	0×3	0
001	0×2	0
010	0×2	0
100	0×2	0
011	1×2	1
101	1×2	1
110	1×2	1
111	1×3	1



2-bit errors are detectable, but not always correctable.

Performance

- Detects any two-bit flips
- Corrects any single-bit flip

Cost

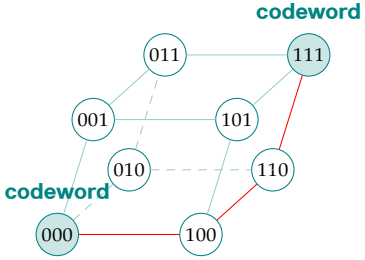
- Rate $R = 1/3$: send 3 bits to convey 1 bit
- Very inefficient!

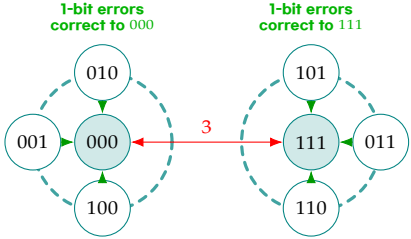
Question. Can we do much better?

I. Introduction
The Key Idea: Distance



Why does the triple repetition code work? The two codewords are **far apart**.





Minimum distance

A code with minimum distance d corrects all errors of weight $\leq \left\lfloor \frac{(d-1)}{2} \right\rfloor$.

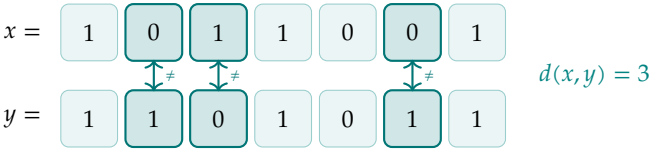
I. Introduction
Hamming Distance



Hamming distance counts how many coordinates differ. For any $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^n$,

$$d_H(\mathbf{x}, \mathbf{y}) = w_H(\mathbf{x} \oplus \mathbf{y}) = \#\{i : x_i \neq y_i\}$$

where w_H is Hamming weight (number of ones).

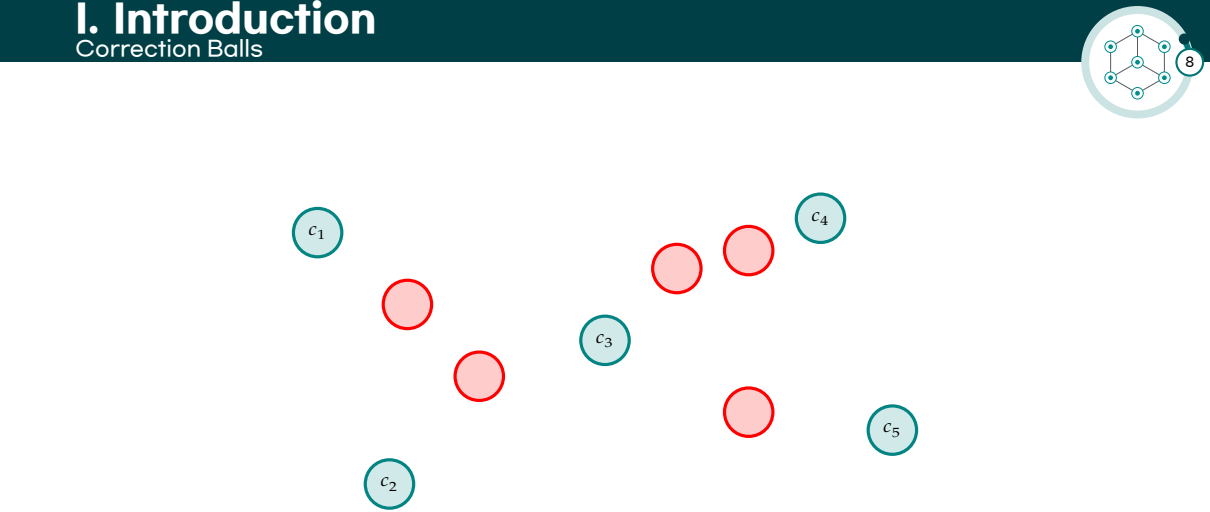


Example 1. $d_H(0101, 1100) = 2$.

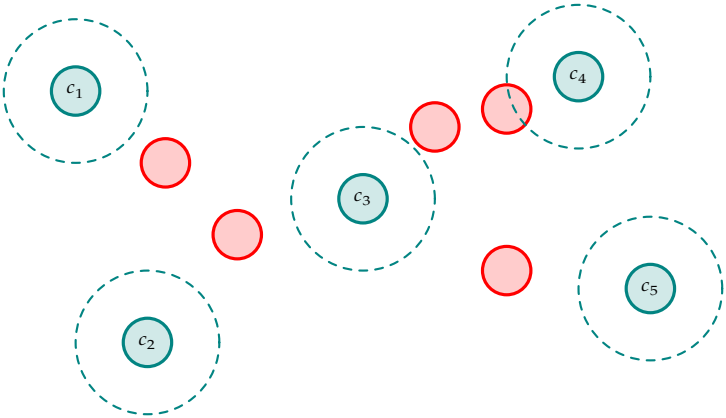
Example 2. $d_H(000, 101) = 2$.

Hamming distance is a metric.

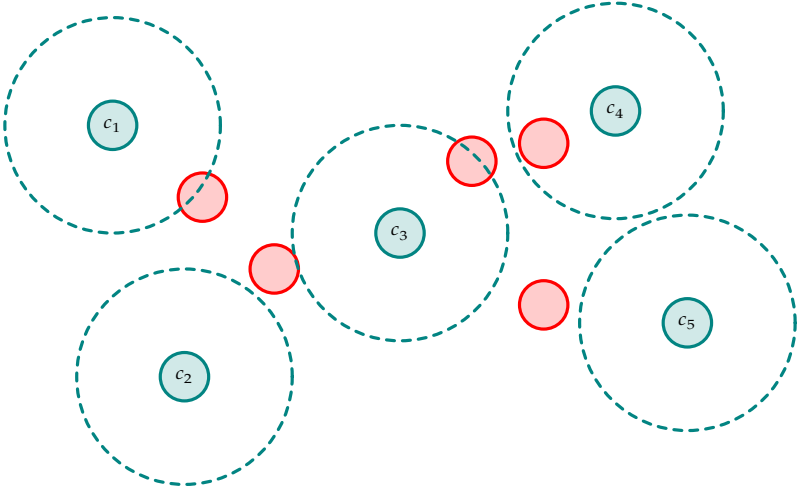
I. Introduction
Correction Balls



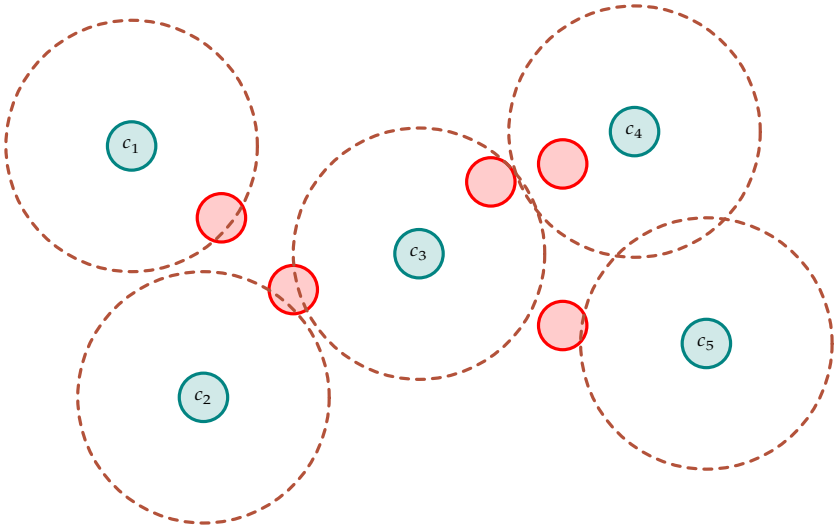
I. Introduction
Correction Balls



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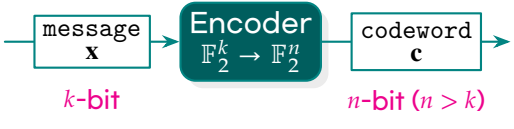


I. Introduction
Correction Balls



Remark. **No overlap of correction balls** $\iff$ unique decoding is possible.

I. Introduction
The Communication Model



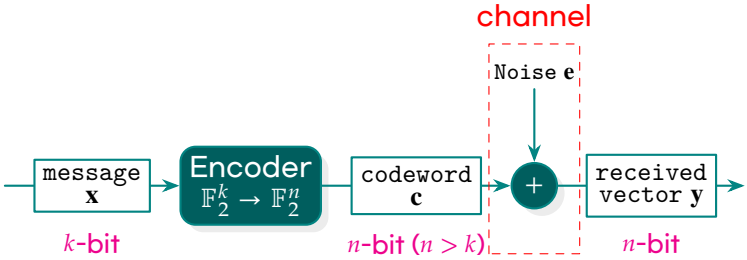
Setup

- Message $\mathbf{x} \in \mathbb{F}_q^k$
- Encoder : $\mathbf{x} \mapsto \mathbf{c} \in \mathbb{F}_q^n,\ n > k$
- Channel : $\mathbf{y} = \mathbf{c} \oplus \mathbf{e}$

Goal

- Recover $\hat{\mathbf{x}} = \mathbf{x}$ even when $\mathbf{e} \neq \mathbf{0}$
- Keep rate $R = k/n$ as large as possible

I. Introduction
The Communication Model



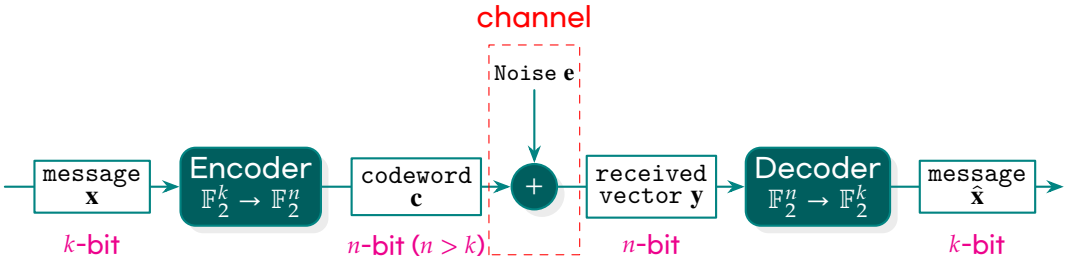
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I. Introduction
The Fundamental Tradeoff



Code	**Rate**	**Detects**	**Corrects** t
$[n,k,d]_q$-code	k/n	$d-1$	$\lfloor (d-1)/2 \rfloor$
Repetition $[3,1,3]_2$	$1/3 \approx 0.33$	2	1
Hamming $[7,4,3]_2$	$4/7 \approx 0.57$	2	1
BCH $[15,7,5]_2$	$7/15 \approx 0.47$	4	2
Reed-Solomon $[15,11,5]_{16}$	$11/15 \approx 0.73$	4	2
Reed-Solomon $[255,223,33]_{256}$	$223/255 \approx 0.87$	32	16

- high rate $\implies$ efficiency
- correct many errors $\implies$ powerful

I. Introduction
Roadmap



II. Linear Codes

Structure via **linear algebra**

- Subspaces of $\mathbb{F}_q^n$
- Generator Matrix G, parity-check H
- Hamming distance & error correction

IV. Cyclic Codes

Structure via **polynomial rings**

- Quotient $\mathbb{F}_q[X]/\langle X^n - 1\rangle$
- Ideals and generator polynomials
- BCH design and cyclic Hamming codes

II. Linear Codes

II. Linear Codes
Linear Code



Linear Encoding

Definition. A **linear encoding** over $\mathbb{F}_q$ is a linear map

$$\mathbb{F}_q^k \longrightarrow \mathbb{F}_q^n, \qquad \mathbf{x} \longmapsto \mathbf{c}.$$

Linear code

Definition. A **linear** $[n,k,d]_q$ **code** is a k-dimensional subspace

$$C \leq \mathbb{F}_q^n$$

with minimum distance d. The elements of C are called **codewords**.

II. Linear Codes
Generator Matrices



Generator matrix

Definition. A **generator matrix** for an $[n,k]_q$ code is a matrix $\mathbf{G} \in \mathbb{F}_q^{k\times n}$ whose rows form a basis of C. Encoding is the linear map

$$\mathbb{F}_q^k \longrightarrow \mathbb{F}_q^n, \qquad \mathbf{m} \longmapsto \mathbf{mG}.$$

Example (The Hamming Code $[7,4,3]_2$). Consider

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}_{4\times 7}$$

Encode message $\mathbf{m} = (1,0,1,1)$:

$$\mathbf{c} = \mathbf{mG} = \begin{bmatrix} 1 & 0 & 1 & 1 \end{bmatrix} \cdot \mathbf{G} = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}.$$

II. Linear Codes
Systematic Form

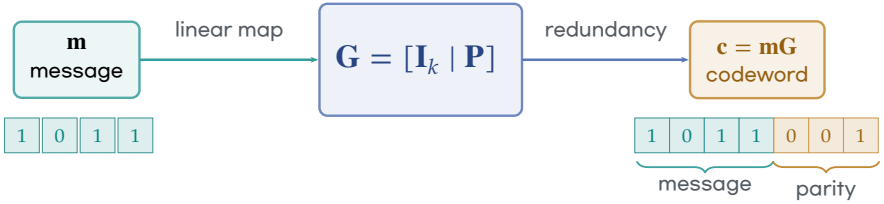


Systematic Form

Definition. A code is said to be in **systematic form** if it has a generator matrix of the form

$$\mathbf{G} = [\mathbf{I}_k \mid \mathbf{A}]$$

- The message appears visibly in the first k coordinates.
- The remaining $n - k$ coordinates are parity symbols.
- Systematic form is convenient, but it is not an intrinsic property of the code.



II. Linear Codes
Parity-check Matrices

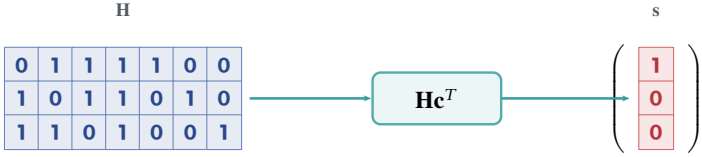


Parity-check Matrix

A **parity-check matrix** is a matrix $\mathbf{H} \in \mathbb{F}_q^{(n-k)\times n}$ such that

$$C = \ker(\mathbf{H}) = \{\mathbf{c} \in \mathbb{F}_q^n : \mathbf{H}\mathbf{c}^T = 0\}.$$

- If $\mathbf{G}$ generates C, then $\mathbf{G}\mathbf{H}^T = 0$.



II. Linear Codes
Duality



Dual code

Definition. The dual of $C \leq \mathbb{F}_q^n$ is

$$C^\perp = \{\mathbf{y} \in \mathbb{F}_q^n : \langle \mathbf{x}, \mathbf{y} \rangle = 0 \text{ for all } \mathbf{x} \in C\}.$$

Remark (Dimension identity). $\dim C + \dim C^\perp = n$.

II. Linear Codes
Syndromes



Syndrome

Proposition. For a received word $\mathbf{y} = \mathbf{c} + \mathbf{e}$,

$$\mathbf{H}\mathbf{y}^T = \mathbf{H}\mathbf{c}^T + \mathbf{H}\mathbf{e}^T = \mathbf{H}\mathbf{e}^T.$$

- $\mathbf{H}\mathbf{y}^T = 0$ if and only if $\vec{y} \in C$.

II. Linear Codes
Correction and detection



Distance theorem

A code of minimum distance d detects every error of weight at most $d-1$ and corrects every error of weight at most

$$t = \left\lfloor \frac{d-1}{2} \right\rfloor.$$

- Detection: no nonzero error of weight $< d$ is a codeword.
- Correction: balls of radius t around distinct codewords are disjoint.

II. Linear Codes
Packing



Hamming bound

If C is an $[n,k,d]_q$ code and $t = \lfloor (d-1)/2 \rfloor$, then

$$q^k \sum_{i=0}^{t} \binom{n}{i} (q-1)^i \leq q^n.$$

II. Linear Codes
Singleton bound



Singleton bound

Every $[n,k,d]_q$ code satisfies

$$d \leq n - k + 1.$$

- Codes meeting the bound are called MDS codes.
- Reed-Solomon codes are the standard nonbinary MDS examples.

II. Linear Codes
Gilbert-Varshamov idea



Existence principle

There exist linear codes whose parameters are good because a greedy construction cannot cover the whole space too early.

$$q^k \sum_{i=0}^{d-2} \binom{n-1}{i} (q-1)^i < q^n \quad \Rightarrow \quad \text{an } [n,k,d]_q \text{ code exists.}$$

II. Linear Codes
Griesmer bound



Griesmer bound

For a linear $[n,k,d]_q$ code,

$$n \geq \sum_{i=0}^{k-1} \left\lceil \frac{d}{q^i} \right\rceil .$$

II. Linear Codes
Weight enumerators



Weight enumerator

The weight enumerator is

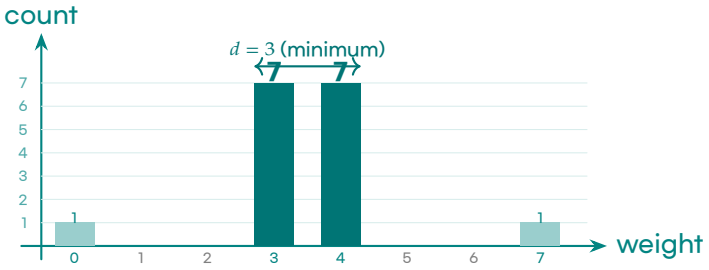
$$W_C(X,Y) = \sum_{c \in C} X^{n-\mathrm{wt}(c)} Y^{\mathrm{wt}(c)}.$$

- It records the distribution of weights.
- The minimum distance is the first nonzero exponent of Y after the constant term.
- It is a coarser invariant than the code but a powerful one.

II. Linear Codes

Weight Distribution of the $[7,4,3]$ Code

The weight enumerator of $\mathrm{Ham}(3,2)$ is $W(X,Y) = X^7 + 7X^4Y^3 + 7X^3Y^4 + Y^7$. This means the 16 codewords have weights:



What we learn

- No codeword has weight 1 or 2 $\Rightarrow d \geq 3$
- The 7 weight-3 words are the Fano plane lines
- Symmetric: C contains the all-ones word

MacWilliams duality

The weight distribution of the dual (simplex) code follows by the MacWilliams transform applied to this bar chart.

II. Linear Codes
MacWilliams identity



MacWilliams identity

For a linear $[n,k]_q$ code,

$$W_{C^\perp}(X,Y) = \frac{1}{|C|} W_C(X + (q-1)Y, X - Y).$$

- The identity is a finite Fourier transform.
- It turns the weight distribution of C into that of $C^\perp$.
- It is one of the central reasons duality is computationally useful.

II. Linear Codes
Self-orthogonality



Self-orthogonal and self-dual

A code is self-orthogonal if $C \subseteq C^{\perp}$ and self-dual if $C = C^{\perp}$.

- Self-dual binary codes have dimension $n/2$.
- Doubly-even self-dual codes have all weights divisible by 4.
- Such codes connect coding theory to lattices and modular forms.

II. Linear Codes
Equivalent codes



Monomial equivalence

Two q-ary linear codes are equivalent if one is obtained from the other by permuting coordinates and multiplying coordinates by nonzero scalars.

- Over $\mathbb{F}_2$, this is just coordinate permutation.
- Equivalent codes have the same parameters and weight enumerator.
- Classification means classification up to this equivalence.

II. Linear Codes
Automorphisms



Automorphism group

$$\mathrm{Aut}(C) = \{\pi \in S_n : \pi(C) = C\}$$

in the binary case.

- A large automorphism group is a sign of hidden geometry.
- Symmetry reduces decoding complexity.
- Hamming and cyclic codes both arise from strong symmetry.

II. Linear Codes
Repetition code



Repetition code

$$C = \{(a, a, \dots, a) : a \in \mathbb{F}_q\}$$

is an $[n, 1, n]_q$ code.

- It has maximal distance for dimension 1.
- Its rate is $1/n$, so it is inefficient.
- It corrects $\lfloor (n-1)/2 \rfloor$ errors by majority vote when $q = 2$.

II. Linear Codes
Single parity-check code



Parity-check code

$$C = \{x \in \mathbb{F}_q^n : x_1 + \cdots + x_n = 0\}$$

is an $[n, n-1, 2]_q$ code.

- The parity-check matrix is $H = [1\ 1\ \cdots\ 1]$.
- It detects one error but cannot correct one error.
- It is the dual of the repetition code.

II. Linear Codes
The simplex code



Binary simplex code

Let H_r be the matrix whose columns are all nonzero vectors of $\mathbb{F}_2^r$. The row space of H_r is the simplex code.

- It has parameters $[2^r - 1, r, 2^{r-1}]_2$.
- Every nonzero codeword has the same weight.
- Its dual is the binary Hamming code.

II. Linear Codes
Matroid viewpoint



Vector matroid of a code

A parity-check matrix H defines a matroid on the coordinate set $\{1, \dots, n\}$ by linear dependence of columns.

- Circuits of the matroid correspond to minimal support codewords.
- The minimum distance is the size of the smallest circuit.
- Projective geometry enters when the columns of H are projective points.

II. Linear Codes
Tanner graphs



Tanner graph

Given a parity-check matrix H, form a bipartite graph with variable nodes $1, \dots, n$ and check nodes equal to the rows of H.

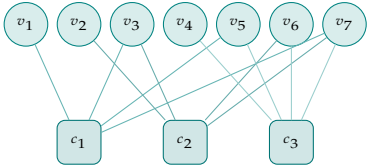
- An edge joins variable j to check i when $H_{ij} \neq 0$.
- Sparse H gives iterative decoding algorithms.
- Classical algebraic codes often have dense checks; LDPC codes emphasize sparsity.

II. Linear Codes

Tanner Graph for the $[7, 4, 3]$ Code



The parity-check matrix H_3 gives a bipartite graph with 7 **variable nodes** and 3 **check nodes**.



How to read the graph

- Variable node v_j = bit j of the codeword
- Check node c_i = parity equation i
- Edge (v_j, c_i) means bit j appears in check i

Each check node says:

$$c_1 : v_1 \oplus v_3 \oplus v_5 \oplus v_7 = 0$$

$$c_2 : v_2 \oplus v_3 \oplus v_6 \oplus v_7 = 0$$

$$c_3 : v_4 \oplus v_5 \oplus v_6 \oplus v_7 = 0$$

II. Linear Codes
Decoding hardness



Syndrome decoding problem

Given H, a syndrome s, and an integer t, decide whether there exists e with

$$He^T = s, \qquad \mathrm{wt}(e) \le t.$$

- This problem is NP-complete in general.
- Code-based cryptography is built around this hardness.
- Algebraic code families are useful because they add trapdoor-like structure.

II. Linear Codes
Perfect codes



Perfect code

A code correcting t errors is perfect if the radius-t balls around codewords partition $\mathbb{F}_q^n$.

$$|C| \sum_{i=0}^{t} \binom{n}{i} (q-1)^i = q^n.$$

- Perfect codes are extremely rare.
- The binary Hamming codes are perfect one-error-correcting codes.
- The Golay codes are the exceptional perfect codes.

II. Linear Codes

Why linear codes are central



Algebra

- subspaces
- duality
- matrices
- finite fields

Algorithms

- encoding by G
- checking by H
- syndrome decoding
- structured families

II. Linear Codes
The road to Hamming



Question

Can we construct a linear code that corrects one error and wastes no syndrome?

- One error at one of n positions gives n possibilities.
- No error gives one more possibility.
- A parity-check matrix with r rows has 2^r binary syndromes.

$$n + 1 = 2^r$$

is the numerology of binary Hamming codes.

III. Hamming Codes

III. Hamming Codes

Hamming (1950): The Original Problem



Richard Hamming worked at Bell Labs in 1947. The relay computer he used ran **punched-card jobs** overnight — and a single wrong hole would crash the job.

His frustration: "If the machine can detect an error, why can't it fix it?"

Hamming's insight (1950)

Add redundant parity bits so that:

- the syndrome identifies the **exact position** of the error, and
- **no syndrome is wasted** — each possible syndrome maps to exactly one error pattern.

This requires $n + 1 = 2^r$ syndromes for n positions, giving $n = 2^r - 1$.

III. Hamming Codes

The design problem



Goal

Construct binary linear codes that correct exactly one error and use all possible syndromes.

- A received word has either no error or one error in position i.
- Thus there are $1 + n$ elementary cases.
- With r parity checks, there are 2^r syndromes.

$$1 + n = 2^r \quad \Rightarrow \quad n = 2^r - 1.$$

III. Hamming Codes

The parity-check matrix



Binary Hamming matrix

For $r \geq 2$, let H_r be the $r \times (2^r - 1)$ matrix whose columns are all nonzero vectors of $\mathbb{F}_2^r$.

$$H_3 = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

- Column labels are nonzero binary syndromes.
- Every nonzero syndrome identifies a unique column.

III. Hamming Codes

Definition



Binary Hamming code

The binary Hamming code of order r is

$$\mathrm{Ham}(r,2) = \ker(H_r) \leq \mathbb{F}_2^{2^r-1}.$$

- Length: $n = 2^r - 1$.
- Redundancy: r.
- Dimension: $k = n - r = 2^r - 1 - r$.

$\mathrm{Ham}(r,2)$ has parameters $[2^r - 1, 2^r - 1 - r, 3]_2$.

III. Hamming Codes

Dimension



Dimension count

The matrix H_r has rank r, hence

$$\dim \ker(H_r) = 2^r - 1 - r.$$

III. Hamming Codes

Minimum distance



Distance

The binary Hamming code $\mathrm{Ham}(r,2)$ has minimum distance 3.

- Therefore Hamming codes correct one error.
- They detect two errors.

III. Hamming Codes

Perfectness



Hamming codes are perfect

The binary Hamming code satisfies equality in the Hamming bound for $t = 1$.

$$2^k(1+n) = 2^{2^r-1-r}(1+2^r-1) = 2^{2^r-1} = 2^n.$$

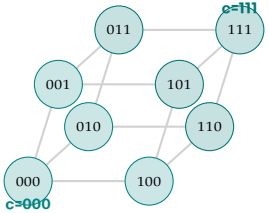
- Every word of $\mathbb{F}_2^n$ is at distance at most 1 from exactly one codeword.
- The syndrome is not merely a diagnostic: it is a complete decoder.

III. Hamming Codes

Perfect Tiling: A Geometric View



For the $[3,1,3]$ repetition code in $\mathbb{F}_2^3$: the two correction balls of radius 1 **tile the cube exactly** — no overlap, no gap.



Why "perfect"?

- Each ball has 4 elements: 1 codeword + 3 neighbors
- Two balls cover all $2^3 = 8$ points of $\mathbb{F}_2^3$
- $2 \times 4 = 8$ — the Hamming bound holds with equality

Hamming codes are the only linear perfect 1-error-correcting codes (apart from trivial cases).

III. Hamming Codes

Syndrome decoding



Decoder

Given $v \in \mathbb{F}_2^n$, compute $s = H_r v^T$.

- If $s = 0$, accept v.
- If $s \neq 0$, find the unique column $H_i = s$.
- Correct by flipping coordinate i.

III. Hamming Codes

The Complete Syndrome Table for $[7,4,3]$



With H_3 as written, the syndrome $(s_1,s_2,s_3)^T$ equals column i when bit i is flipped.

Syndrome	Error pattern e	Action
$(0,0,0)$	0000000	no error — accept
$(1,0,0)$	1000000	flip bit 1
$(0,1,0)$	0100000	flip bit 2
$(1,1,0)$	0010000	flip bit 3
$(0,0,1)$	0001000	flip bit 4
$(1,0,1)$	0000100	flip bit 5
$(0,1,1)$	0000010	flip bit 6
$(1,1,1)$	0000001	flip bit 7

- All $2^3 = 8$ syndromes are used — confirming **perfectness**
- Decoding is a single table look-up — $O(1)$ hardware

III. Hamming Codes

Binary index convention



Indexed columns

If the i-th column of H_r is the binary expansion of i, then the nonzero syndrome is literally the error position.

$$s = (1,0,1)^T \quad \rightsquigarrow \quad i = 5$$

depending on the chosen bit order.

- This is a notational convention, not an invariant.
- It explains the classical parity-bit positions $1, 2, 4, \ldots$.

III. Hamming Codes

The classical [7,4,3] code



$$H = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

- There are 7 nonzero syndromes.
- Each syndrome names one error coordinate.
- The kernel has dimension 4.

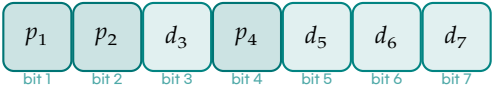
$$\mathrm{Ham}(3,2) = [7,4,3]_2.$$

III. Hamming Codes

The $[7,4,3]$ Code: Parity Bit Positions



Bits are numbered 1 through 7. Positions that are powers of 2 $(1,2,4)$ are **parity bits**; the rest carry **data**.



Parity check coverage:

Parity bit	Checks bits	Syndrome row
p_1 (bit 1)	1, 3, 5, 7	H row 1: $(1,0,1,0,1,0,1)$
p_2 (bit 2)	2, 3, 6, 7	H row 2: $(0,1,1,0,0,1,1)$
p_4 (bit 4)	4, 5, 6, 7	H row 3: $(0,0,0,1,1,1,1)$

- Each parity bit checks all positions whose index has a 1 in the corresponding binary place
- The 3-bit syndrome tells us the binary address of the flipped bit

III. Hamming Codes

A systematic generator matrix



One systematic form for the $[7,4,3]$ Hamming code is

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

- The first four coordinates are message symbols.
- The last three are parity symbols.
- This G is equivalent to the kernel of a suitable H.

III. Hamming Codes

Encoding example



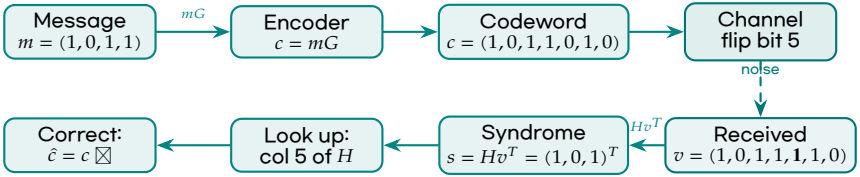
For $m = (1,0,1,1)$ and the systematic matrix G,

$$c = mG = (1,0,1,1,0,0,1).$$

- The parity bits are linear forms in the message bits.
- A single flipped bit changes the syndrome to the corresponding column.
- The correction rule does not require knowing the transmitted message.

III. Hamming Codes

Encoding & Error Correction



Key numbers:

- Syndrome $s = (1,0,1)^T$ matches column 5 of H_3
- Flip bit 5: $(1,0,1,1,\mathbf{1},1,0) \to (1,0,1,1,0,1,0)$
- Recovered $\hat{c}$ is the original codeword (v)

The decoder never needs m
The syndrome alone identifies the error. Once we flip the right bit, the first 4 bits of the corrected $\hat{c}$ directly give $\hat{m} = (1,0,1,1)$.

III. Hamming Codes

Projective geometry



Projective interpretation

The nonzero columns of H_r, modulo scalar multiplication, are the points of projective space

$$\mathbb{P}^{r-1}(\mathbb{F}_q).$$

- Over $\mathbb{F}_2$, each nonzero vector is its own projective point.
- Over $\mathbb{F}_q$, scalar multiples define the same projective point.
- Hamming codes are parity checks indexed by projective points.

III. Hamming Codes

The Fano plane



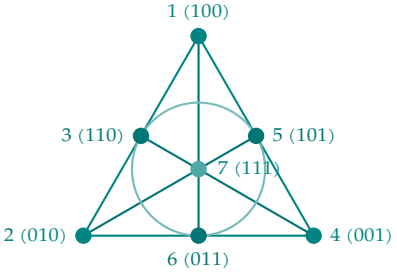
For $r = 3$, the projective space $\mathbb{P}^2(\mathbb{F}_2)$ has 7 points and 7 lines.

- The 7 columns of H_3 are the points.
- A dependent triple of columns is a line of the Fano plane.
- Weight-3 codewords are exactly these lines.

The minimum distance 3 is the statement that the smallest projective line has three points.

III. Hamming Codes

The Fano Plane — Drawn



The 7 lines (each has 3 points):

#	Points	Columns sum to 0
1	$\{1,2,3\}$	100+010=110
2	$\{1,4,5\}$	100+001=101
3	$\{2,4,6\}$	010+001=011
4	$\{1,6,7\}$	100+011=111
5	$\{2,5,7\}$	010+101=111
6	$\{3,4,7\}$	110+001=111
7	$\{3,5,6\}$	110+101=011

III. Hamming Codes

Lines give codewords



In $\mathbb{P}^2(\mathbb{F}_2)$, if a, b, c are the three points on a line, then

$$a + b + c = 0.$$

- The incidence vector of that line is a Hamming codeword.
- The 7 lines give the 7 words of weight 3.
- Their complements give 7 words of weight 4.

$$W_{\mathrm{Ham}(3,2)}(Y) = 1 + 7Y^3 + 7Y^4 + Y^7.$$

III. Hamming Codes

The simplex dual



Simplex code

The row space of H_r is the binary simplex code:

$$\mathrm{Sim}(r,2) = \mathrm{rowspan}(H_r).$$

- It has parameters $[2^r - 1, r, 2^{r-1}]_2$.
- Every nonzero word has weight 2^{r-1}.
- It is dual to the Hamming code.

$$\mathrm{Ham}(r,2)^{\perp} = \mathrm{Sim}(r,2).$$

III. Hamming Codes

MacWilliams check



For $r = 3$, the simplex code has weight enumerator

$$W_{\mathrm{Sim}}(X, Y) = X^7 + 7X^3Y^4.$$

Applying MacWilliams gives

$$W_{\mathrm{Ham}}(X, Y) = X^7 + 7X^4Y^3 + 7X^3Y^4 + Y^7.$$

- The algebraic transform recovers the geometric weight distribution.
- Duality is visible both in matrices and in enumerators.

III. Hamming Codes

Q-ary Hamming codes



Q-ary construction

Choose one nonzero representative from each one-dimensional subspace of $\mathbb{F}_q^r$ and use these vectors as columns of H.

$$n = \frac{q^r - 1}{q - 1}.$$

- The resulting code has parameters

$$\left[\frac{q^r - 1}{q - 1}, \frac{q^r - 1}{q - 1} - r, 3\right]_q.$$

- It is again perfect for one-error correction.

III. Hamming Codes

Why projective representatives



Over $\mathbb{F}_q$, two proportional nonzero columns are linearly dependent.

- To have $d \geq 3$, no two columns may be proportional.
- Therefore columns must be projective points, not raw nonzero vectors.
- Taking all projective points uses the maximum possible number of columns.

Hamming codes are the maximal parity-check configurations with no zero column and no repeated projective point.

III. Hamming Codes

Q-ary perfectness



For the q-ary Hamming code,

$$n = \frac{q^r - 1}{q - 1}, \qquad k = n - r.$$

The radius-one sphere has size

$$1 + n(q-1) = 1 + q^r - 1 = q^r.$$

Hence

$$q^k(1 + n(q-1)) = q^{n-r}q^r = q^n.$$

Every received word lies at distance at most one from a unique codeword.

III. Hamming Codes

Extended Hamming codes



Extension

Add one overall parity coordinate to force even total weight.

- The binary $[7,4,3]$ code becomes an $[8,4,4]$ code.
- The new minimum distance is 4.
- It detects three errors and corrects one error.

$$\widehat{\mathrm{Ham}}(3,2) = [8,4,4]_2.$$

III. Hamming Codes

The [8,4,4] code



The extended Hamming code has weight enumerator

$$W(X,Y) = X^8 + 14X^4Y^4 + Y^8.$$

- All codeword weights are divisible by 4.
- The code is self-dual.
- It is the smallest doubly-even self-dual binary code.

This code is the length-8 seed behind the E_8 lattice construction.

III. Hamming Codes

Automorphisms



Symmetry group

The automorphism group of the binary Hamming code is induced by invertible linear transformations of $\mathbb{F}_2^r$:

$$\mathrm{Aut}(\mathrm{Ham}(r,2)) \cong \mathrm{GL}(r,2).$$

- A matrix in $\mathrm{GL}(r,2)$ permutes the nonzero columns of H_r.
- Hence it permutes coordinates and preserves the kernel.
- The code remembers the projective geometry.

III. Hamming Codes

Tanner graph



- The Tanner graph of H_r has $n = 2^r - 1$ variable nodes and r check nodes.
- It is dense compared with modern LDPC codes.
- But it has an exceptionally simple exact decoder.

Interpretation

For Hamming codes, the syndrome is not iteratively refined; it is read directly as a projective point.

III. Hamming Codes

Failure modes



- A two-error pattern has syndrome equal to the sum of two columns.
- In a binary Hamming matrix, that sum is another nonzero column.
- Therefore a two-error pattern may be miscorrected as a one-error pattern.

Minimum distance 3 guarantees single-error correction, not reliable detection after attempting correction.

III. Hamming Codes

Hamming codes as BCH codes



Cyclic viewpoint

When $n = 2^r - 1$, the binary Hamming code can be realized as a narrow-sense BCH code of designed distance 3.

- Choose a primitive n-th root $\alpha \in \mathbb{F}_{2^r}$.
- Let $g(X)$ be the minimal polynomial of α over $\mathbb{F}_2$.
- Then the cyclic code generated by $g(X)$ is a Hamming code.

III. Hamming Codes

The [7,4,3] cyclic form



Over $\mathbb{F}_2$,

$$X^7 - 1 = (X+1)(X^3 + X + 1)(X^3 + X^2 + 1).$$

Taking

$$g(X) = X^3 + X + 1$$

gives a cyclic $[7, 4, 3]$ Hamming code.

- This is the bridge from projective checks to polynomial ideals.
- The next chapter makes that bridge systematic.

III. Hamming Codes

Classification intuition



- Perfect one-error-correcting linear codes are essentially Hamming codes, apart from trivial cases.
- The parity-check matrix must use every possible nonzero syndrome once.
- Thus the construction is forced by the Hamming bound.

Hamming codes are beautiful because optimality, decoding, and geometry are the same statement.

III. Hamming Codes

Summary



Construction

- parity-check columns are projective points
- no repeated projective column
- all syndromes are used

Consequences

- parameters $[2^r - 1, 2^r - 1 - r, 3]_2$
- perfect one-error correction
- dual simplex code

III. Hamming Codes

Transition to cyclic codes



New symmetry

Hamming codes are governed by projective geometry. Cyclic codes are governed by the cyclic shift operator.

- Linear code: subspace of $\mathbb{F}_q^n$.
- Hamming code: subspace defined by projective parity checks.
- Cyclic code: subspace stable under a cyclic permutation.

$$\text{shift symmetry} \iff \text{ideals in } \mathbb{F}_q[X]/(X^n - 1).$$

IV. Cyclic Codes

IV. Cyclic Codes

Cyclic shift



Cyclic code

A linear code $C \leq \mathbb{F}_q^n$ is cyclic if

$$(c_0, c_1, \dots, c_{n-1}) \in C \quad \Rightarrow \quad (c_{n-1}, c_0, \dots, c_{n-2}) \in C.$$

- This is invariance under a single permutation.
- Since the permutation has order n, cyclic codes are modules over a cyclic group algebra.
- The polynomial model makes this precise.

IV. Cyclic Codes

Polynomial dictionary



Vector to polynomial

Identify

$$(c_0, c_1, \dots, c_{n-1}) \quad \longleftrightarrow \quad c(X) = c_0 + c_1 X + \cdots + c_{n-1} X^{n-1}.$$

- Multiplication by X shifts coefficients.
- The relation $X^n = 1$ performs wrap-around.
- Therefore we work modulo $X^n - 1$.

IV. Cyclic Codes

The quotient ring



Ambient ring

$$R_n = \mathbb{F}_q[X]/(X^n - 1).$$

- Vectors of length n are polynomials of degree $< n$.
- Cyclic shift is multiplication by X in R_n.
- A cyclic code is an $\mathbb{F}_q$-subspace stable under multiplication by X.

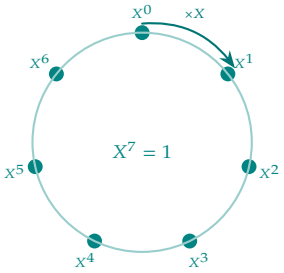
Which subspaces of R_n are stable under multiplication by X?

IV. Cyclic Codes

R_n: The Ring as a Rotation

Think of $R_n = \mathbb{F}_q[X]/(X^n - 1)$ like a **clock with n positions** $0, 1, \ldots, n-1$.

Multiplying by X = **rotating one step**.



From integers to polynomials

$\mathbb{Z}/n\mathbb{Z}$	$\mathbb{F}_q[X]/(X^n-1)$
add n to get 0	$X^n = 1$
$+1$ = shift right	$\times X$ = shift
subgroup = ideal	subspace closed under $\times X$ = ideal

Why this matters

A cyclic code is a subspace of R_n that is closed under rotation. That is exactly an **ideal** — generated by a single divisor $g(X)$ of $X^n - 1$.

IV. Cyclic Codes

Ideals



Cyclic codes are ideals

Under the identification $\mathbb{F}_q^n \cong R_n$, cyclic codes are exactly the ideals of R_n.

IV. Cyclic Codes

Principal ideals



Principal generator

Every ideal of R_n is generated by a unique monic divisor $g(X)$ of $X^n - 1$:

$$C = (g(X)).$$

- This follows from the principal ideal property of $\mathbb{F}_q[X]$.
- The generator polynomial is the monic element of least degree in the code.
- The divisibility condition is $g(X) \mid X^n - 1$.

IV. Cyclic Codes

Dimension



Dimension formula

If $C = (g(X)) \subseteq R_n$ and $\deg g = r$, then

$$\dim_{\mathbb{F}_q} C = n - r.$$

IV. Cyclic Codes

Generator matrix from shifts



If

$$g(X) = g_0 + g_1 X + \cdots + g_r X^r,$$

then a generator matrix is obtained by shifting the coefficient vector:

$$\begin{bmatrix} g_0 & g_1 & \cdots & g_r & 0 & \cdots & 0 \\ 0 & g_0 & g_1 & \cdots & g_r & \cdots & 0 \\ \vdots & & \ddots & & & \ddots & \vdots \end{bmatrix}.$$

- This matrix is Toeplitz-like.
- It encodes multiplication by $g(X)$.

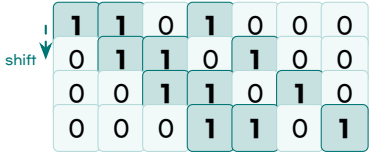
IV. Cyclic Codes

The Shifted (Toeplitz) Structure of the Generator Matrix



For $g(X) = X^3 + X + 1$ and the $[7,4,3]$ cyclic code, the generator matrix is:

$$G = \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} & 0 & 0 & 0 \\ 0 & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} & 0 \\ 0 & 0 & 0 & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{1} \end{bmatrix}$$



- Each row is a **one-position cyclic shift** of the row above
- The non-zero diagonals follow the pattern of g's coefficients $(1,1,0,1)$
- This "staircase" pattern is what makes encoding a shift-register operation

IV. Cyclic Codes

Parity-check polynomial



Check polynomial

If $g(X) \mid X^n - 1$, define

$$h(X) = \frac{X^n - 1}{g(X)}.$$

- $\deg h = k$ when C has dimension k.
- Since $g(X)h(X) = 0$ in R_n, multiplication by $h(X)$ checks membership.
- More precisely, $c(X) \in (g)$ implies $c(X)h(X) \equiv 0 \pmod{X^n - 1}$.

IV. Cyclic Codes

Reciprocal polynomials



Reciprocal

For $f(X) = f_0 + \cdots + f_m X^m$ with $f_0 \neq 0$, define

$$f^*(X) = X^m f(X^{-1}).$$

- Parity-check matrices are naturally written using reciprocal polynomials.
- The dual of a cyclic code is cyclic.
- If $C = (g)$ and $h = (X^n - 1)/g$, then $C^\perp$ is generated by a reciprocal of h.

IV. Cyclic Codes

Idempotents



Idempotent generator

When $\gcd(n,q) = 1$, each cyclic code has a unique idempotent generator $e(X)$ satisfying

$$e(X)^2 = e(X), \qquad C = (e(X)).$$

- Idempotents decompose the semisimple ring R_n.
- They expose the representation-theoretic structure.
- Generator polynomials are usually better for computation.

IV. Cyclic Codes

Roots



Let α be an n-th root of unity in an extension field.

- A cyclic code generated by $g(X)$ has zeros α^i where $g(\alpha^i) = 0$.
- The defining set is
$$Z(C) = \{i \pmod{n} : g(\alpha^i) = 0\}.$$
- Root sets translate algebraic divisibility into combinatorics modulo n.

IV. Cyclic Codes

Cyclotomic cosets



Q-cyclotomic coset

Modulo n, the q-cyclotomic coset of i is

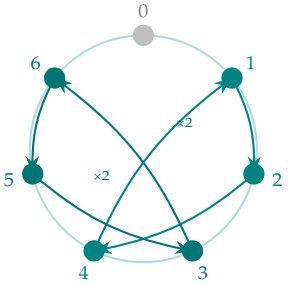
$$C_i = \{i, iq, iq^2, \ldots\} \pmod{n}.$$

- If α^i is a root over $\mathbb{F}_q$, all conjugates α^{iq^j} are roots.
- Therefore defining sets are unions of cyclotomic cosets.
- Minimal polynomials correspond to these cosets.

IV. Cyclic Codes

Cyclotomic Cosets for $n = 7, q = 2$: Orbits Under $\times 2$

Multiplying by $q = 2$ modulo $n = 7$ partitions $\{0, 1, \ldots, 6\}$ into orbits:



Three cosets:

Coset	Elements	Min. poly
C_0	$\{0\}$	$X + 1$
C_1	$\{1, 2, 4\}$	$X^3 + X + 1$
C_3	$\{3, 5, 6\}$	$X^3 + X^2 + 1$

This gives the factorization:

$$X^7 - 1 = \underbrace{(X+1)}_{C_0}\,\underbrace{(X^3+X+1)}_{C_1}\,\underbrace{(X^3+X^2+1)}_{C_3}$$

Cyclic codes = choose a subset of factors as the generator $g(X)$.

IV. Cyclic Codes

Minimal polynomials



Minimal polynomial

For α^i, the minimal polynomial over $\mathbb{F}_q$ is

$$M_i(X) = \prod_{j \in C_i} (X - \alpha^j).$$

- It lies in $\mathbb{F}_q[X]$.
- It is irreducible.
- Generator polynomials of cyclic codes are products of such $M_i(X)$.

IV. Cyclic Codes

BCH codes



BCH design

A BCH code is a cyclic code whose generator polynomial has a prescribed block of consecutive roots.

$$g(\alpha^b) = g(\alpha^{b+1}) = \cdots = g(\alpha^{b+\delta-2}) = 0.$$

- The integer δ is the designed distance.
- Consecutive roots force many syndromes to vanish.
- The BCH bound converts this into a distance guarantee.

IV. Cyclic Codes

BCH bound



BCH bound

If a cyclic code has $\delta - 1$ consecutive roots

$$\alpha^b, \alpha^{b+1}, \dots, \alpha^{b+\delta-2},$$

then its minimum distance is at least δ.

IV. Cyclic Codes

Narrow-sense primitive BCH



Common special case

Let $n = q^m - 1$ and let α be primitive in $\mathbb{F}_{q^m}$. The narrow-sense BCH code of designed distance δ has roots

$$\alpha, \alpha^2, \ldots, \alpha^{\delta-1}.$$

- Primitive: the length is the full multiplicative order.
- Narrow-sense: the first root is α.
- Hamming codes occur when $\delta = 3$ in the binary case.

IV. Cyclic Codes

Cyclic Hamming code



Let $n = 2^r - 1$ and let $\alpha \in \mathbb{F}_{2^r}$ be primitive.

- Let $g(X)$ be the minimal polynomial of α over $\mathbb{F}_2$.
- Then $\deg g = r$.
- The cyclic code (g) has parameters

$$[2^r - 1, 2^r - 1 - r, 3]_2.$$

This is the binary Hamming code in cyclic coordinates.

IV. Cyclic Codes

The case n equals 7



Over $\mathbb{F}_2$,

$$X^7 - 1 = (X+1)(X^3 + X + 1)(X^3 + X^2 + 1).$$

- A primitive 7-th root has minimal polynomial $X^3 + X + 1$ or $X^3 + X^2 + 1$.
- Either cubic gives a cyclic $[7, 4, 3]$ Hamming code.
- The two choices are equivalent under coordinate reversal.

IV. Cyclic Codes

Encoding by multiplication



For a cyclic code (g) with $\deg g = r$, non-systematic encoding is

$$m(X) \longmapsto c(X) = m(X)g(X), \qquad \deg m < n - r.$$

- The message polynomial has $k = n - r$ coefficients.
- The product automatically lies in the ideal (g).
- Hardware can implement this multiplication by shift registers.

IV. Cyclic Codes

Systematic cyclic encoding



Let $\deg g = r$. To encode a message $m(X)$ with $\deg m < k$:

$$X^r m(X) = q(X)g(X) + p(X), \qquad \deg p < r.$$

Then set

$$c(X) = X^r m(X) - p(X).$$

- Since $c(X)$ is divisible by $g(X)$, it is a codeword.
- The high-degree coefficients contain the message.
- The low-degree polynomial $p(X)$ supplies parity.

Thank You

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